

Riddhi
MACHINE TOOL TECHNOLOGY

Engineering confindence in every cut

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COMPANY STATEMENT

Riddhi
MACHINE TOOL TECHNOLOGY

OUR VISION

To be a trusted and reliable one-stop solution for advanced precision cutting tools across automotive, aerospace, and engineering industries.

OUR MISSION

To advance the *"Make in India"* vision by manufacturing world-class precision cutting tools that meet global standards of quality and performance.



TRANSFORMING RAW MATERIALS INTO FINER TOOLS

Riddhi Machine Tool Technology is a precision metal-cutting tools manufacturer delivering solid carbide and PCD solutions through advanced and innovative technologies. The company designs and manufactures tools using state-of-the-art CNC, 5-axis SAACKE grinding, and VOLLMER Wire EDM machines, serving the automotive, aerospace, and precision engineering industries, in line with the vision of *“Make in India.”*




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SAACKE
UW II
VOLLMER
QWD
SAACKE
UW ID
WALTER
HELICHECK PRO

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SOLID CARBIDE



SOLID CARBIDE ENDMILLS

Precision Milling. Consistent Performance.

END MILLS
The high-performance end mills from RIDDHI ensure excellent, reliable results during all machining tasks.
The focus is on cost-effectiveness and productivity, along with meeting customers' demanding requirements.

The application range includes:

- Slot and shoulder milling
- Groove and chamfer milling
- Rough and finish milling
- High-feed milling, 3D profile milling

Irrespective of whether customers need cost-effective milling cutters for all milling applications, RIDDHI offers comprehensive, right solutions based on customers' exact demands, offering specially tailored milling solutions.

END MILL PRODUCT RANGE
Diameter: 2 to 32 mm
No. of flutes / cutting edges (Z): 2, 3, 4, 5, 6
Technology: Multi-helix design with
high wear-resistance coatings
Features: High-polished and edge-honed





SOILD CARBIDE REAMERS

PRECISION REAMING FOR DEMANDING MATERIALS

RIDDHI high-tech and high-performance reamers with internal coolant supply are the ideal solutions for precision bore machining of all kinds of workpiece materials.

We are highly specialised in developing our know-how and have established niche areas in machining precision bores in the automotive and aerospace industries. Our reaming solutions are widely deployed where critical bore tolerances and tight geometrical accuracy are required for workpiece materials like grey cast iron, nodular iron, alloyed steel, and stainless steel, as well as advanced materials like titanium and Inconel.

The multi-blade reamers, together with unequal spacing, specific cutting geometry, and monoblock design, offer a comprehensive full-carbide reamer program for the highest performance.

Product Range:

- Diameter Range: Dia. 2–32 mm
- Tolerance: Customer-specific, e.g., H6, H7, etc.
- Straight flutes for through and blind holes
- LH spiral flutes for through holes
- Ultra-fine carbide material
- Special multi-step reamers
- RIDDHI know-how tool geometry with suitable PVD wear-resistant coating

BENEFITS :

- For very small hole tolerance , and first bore is the right bore !!!
- Extremely high rigidity for optimum performance
- High surface quality and , geometrical tolerance of each bore.



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SOILD CARBIDE DRILLS

PRECISION DRILLING FOR DEMANDING MATERIALS

RIDDHI offers suitable solid carbide drills for practically all workpiece materials, mainly Grey cast iron, Nodular iron, Alloyed steel, Stainless steel, PH Steel, and titanium alloys.

The drills are superficially designed with smooth flutes and effective 2 & 3 edges based on individual applications; cutting edges are edge-honed according to the application.
For aluminium alloy applications, we offer straight-fluted, high-polished, and edge-honed burnishing designs.

We have a specific design suitable for aerospace aluminium alloy because of the low silicon content in this particular material; drilling is always challenging due to long chips and the built-up edge effect on the drilling point cutting edge. We avoid this phenomenon with our specific rake design, together with high gloss polish fluted design, adapting helical flutes together with internal coolant supply.

Product range and technology

- Dia 2-25 mm, Standard as well as special step drills
- Internal coolant supply (dia 4 and above)
- 30 deg standard helix
- 40 deg helix for multi-step tools
- Ultra-fine carbide with suitable wear resistance PVD coating
- Flutes are well polished for better chip flow
- Cutting edges are honed to increase tool life

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POLYCRYSTALLINE DIAMOND

PCD TECHNOLOGY

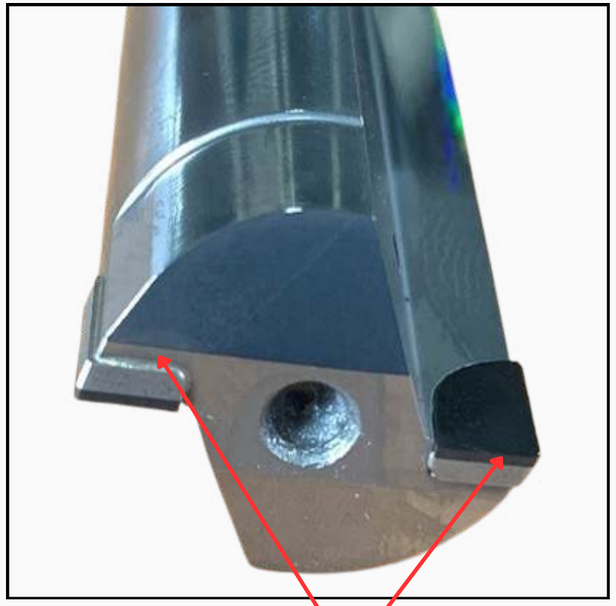


Well balanced PCD material

High quality pocket seats

Perfect smooth flutes

Solid Carbide Body



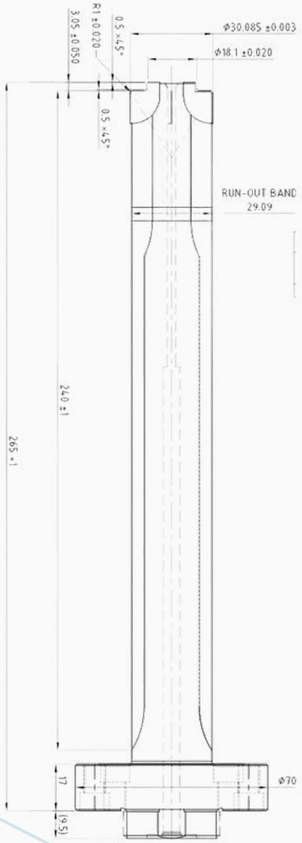
Seamless Brazing

Wire EDM+ Lapping and grinding cutting Edges



Steel Body





PRECISION PCD TOOLING FOR HIGH-PERFORMANCE MACHINING

PCD (Polycrystalline Diamond) tools are the top choice for highly productive machining of aluminium parts in the automotive and aerospace industries.
RIDDHI adapts its extensive experience and know-how, offering application-based individual solutions for drilling, milling, and precision boring PCD tools.

Our approach begins with understanding the customer's component and application requirements. Based on these insights, we provide application-specific engineering solutions, covering the complete PCD tooling process—from proposal and tool design to manufacturing at our state-of-the-art facility—ensuring solutions that meet the customer's performance and production requirements.

Why PCD?
PCD cutting material has the highest hardness of any other cutting material and a very smooth chip surface. Aluminium has an affinity to attract built-up edges at the cutting zone due to its soft nature and the heat generated while cutting, thus sticking quickly to the cutting edge. This is very bad for machining performance, especially if the tools are made with solid carbide material.

PCD, due to its highest hardness together with an ultra-fine microstructure, high surface smoothness, and sharp cutting edges, has zero or very low effect of built-up edge, especially while machining aluminium. This helps to operate the tool at the highest cutting speeds and feeds, reducing machining time drastically together with very long tool life.

Benefits:

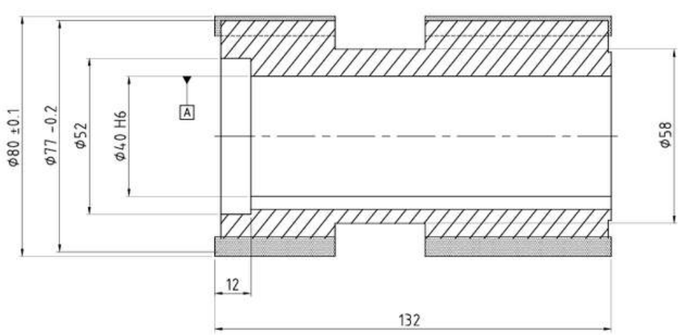
- Highest wear resistance, longer tool life
- Tight tolerances can be achieved for longer machining cycles
- Surface finish less than Ra 0.2 is possible to achieve
- Multi-operations can be combined into a single tool, thus eliminating the need for multiple tools, lowering production costs, and achieving the highest process capability

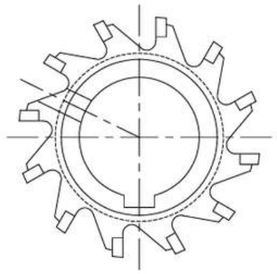
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BRAZED CARBIDE







Solid Performance. Smarter Tooling

Due to the high cost of carbide raw material, RIDDHI brings a unique solution for reducing tool costs while maintaining the same performance as that of solid carbide tools by offering brazed carbide-tipped tools.

The Construction:
The tool body is made of high-quality tool steel, which is heat-treated, and the carbide tips are brazed onto the steel body (suitably fluted and pockets are prepared).
The cutting edges are ground on a 5-axis CNC tool grinding machine, ensuring the required geometries are achieved in one go.

Our Solutions:
Tools are manufactured according to customer-specific requirements
Brazed carbide reamers with internal coolant supply and PVD coating
Stepped special tools and form tools
T-slot milling cutters
Lugged carbide drills and reamers with internal coolant supply

INDUSTRIES WE SERVE

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AUTOMOTIVE





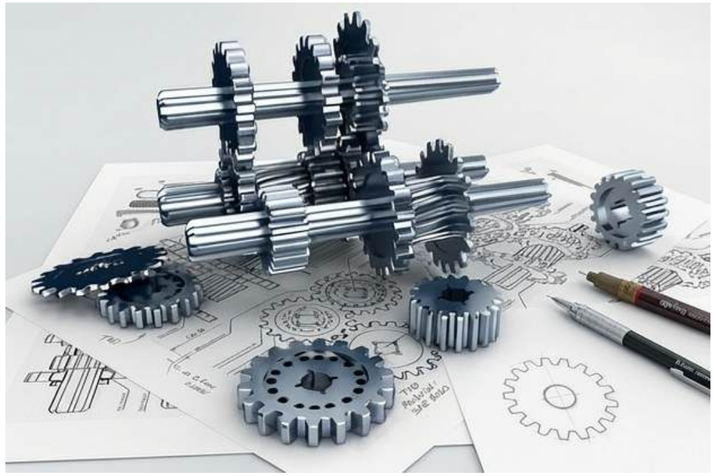
AEROSPACE

DEFENCE

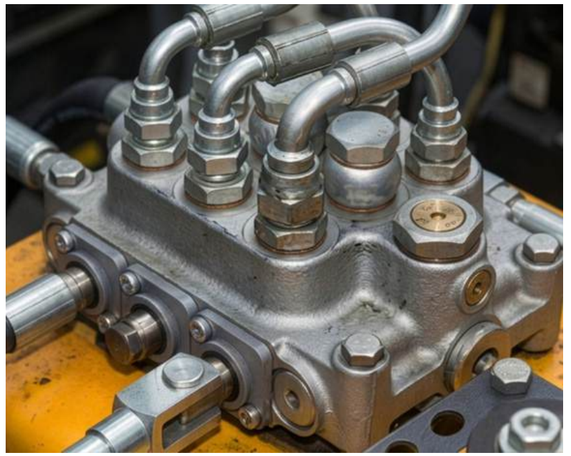




ENERGY



GENERAL ENGINEERING



HYDRAULIC

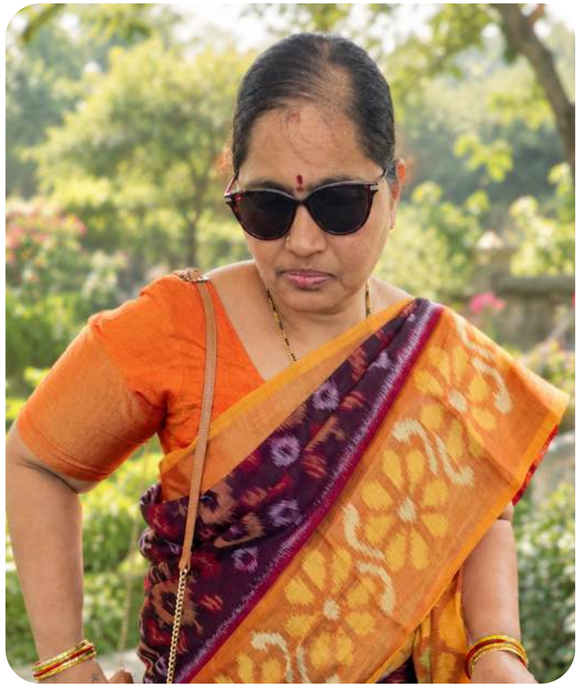
THE VISIONARIES BEHIND OUR JOURNEY



Roshan Tanuk



k.Ravi Kumar



K.Syamala Kumar

Founders note: Riddhi Machine Tool Technology stands for more than just a reliable supplier/manufacturer of high-quality PCD tools & solid carbide tools; it stands for technical expertise, collaboration, partnerships and solutions that make customers more successful in the long term.



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ENGINEERING CONFIDENCE IN EVERY CUT

- Make In India

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THANK YOU